

Exploratory Testing

An approach where learning, test design and execution happen together and are guided by risk.

What is Exploratory Testing?

Exploratory Testing integrates learning, design and execution. Instead of following only previously written cases, QA observes the product, formulates hypotheses, adapts next steps and investigates risks as new information is discovered. Exploring does not mean testing without a method.

When it is valuable

It is especially useful for new features, incomplete requirements, high-risk areas, defect investigations, limited time or when we want to find problems that repeated scripts may not reveal.

Charters

A charter defines a mission without turning the session into a rigid script. Example: “Explore the cart focusing on quantity changes, persistence and price calculation using interruptions and extreme data”.

Heuristics

Use heuristics to vary data, states, sequence, users, permissions, platforms, network, interruptions, time and error conditions. Heuristics help generate questions; they are not mandatory checklists.

Practical example

When exploring image upload: supported/unsupported types, empty file, very large file, long name, special characters, interrupted network, repeated upload, going back during processing and refreshing the page.

Sessions and timebox

A 45–90 minute session with a clear objective helps balance freedom and focus. At the end, debrief: what was tested, discovered, not covered and which questions emerged.

Evidence

Record notes, data, paths, screenshots, videos, logs, defects and ideas. Professional exploration needs to produce communicable learning, even without a detailed test case for every action.

Exploratory vs ad hoc

Ad hoc can be completely improvised. Exploratory Testing is intentional, mission- and learning-oriented. It can use charters, heuristics, mind maps and Session-Based Test Management.

Quick checklist



Mission; risk; timebox; data; heuristics; notes; evidence; defects; questions; uncovered areas; debriefing.

QA Mindset

The value of exploratory testing is learning quickly and using that learning to decide the next test, not clicking randomly.